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Pedestrian Safety Audits near Construction Sites via Open-Vocabulary Vision-Language Segmentation with Few-Shot Filtering

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Abstract:	Ensuring pedestrian safety near construction sites requires frequent audits to detect temporary obstructions, yet current practices are labor-intensive and time-consuming. This study proposes a novel framework that combines open-vocabulary vision-language segmentation with few-shot filtering to automatically generate pseudo-labeled datasets of sidewalk obstructions using street-view images. By integrating Autodistill and a few-shot filtering process, the framework effectively discards unreliable predictions and retains only semantically consistent mask–label pairs. Experimental results show significant improvements in F1 score and MCC compared to baseline models, with especially notable gains for construction domain-specific classes such as fences and traffic cones. The approach also reduces annotation time considerably, enabling rapid dataset generation with minimal human input. As a result, the study contributes a scalable and low-effort approach to auditing construction-related risks to pedestrian safety and offers practical tools for city officials to enhance sidewalk monitoring across diverse urban contexts.

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Aug. 17, 2025

Dr. Daniel Castro-Lacouture
Editor-in-Chief, Automation in Construction

Re: Automation in Construction, New Manuscript

Dear Dr. Castro-Lacouture,

We are pleased to submit our manuscript entitled,

"Pedestrian Safety Audits near Construction Sites via Open-Vocabulary Vision-Language Segmentation with Few-Shot Filtering"
by Taegwan Yoon, Jongwoo Cho, and Seulbi Lee

for consideration in *Automation in Construction*.

This study presents an automated framework that combines open-vocabulary vision-language segmentation with few-shot filtering to generate reliable pseudo-labeled datasets of temporary obstructions in street-view imagery near construction sites.

We believe the findings will be of interest to the journal's readership, including researchers in computer vision-based automated inspection and practitioners seeking intelligent tools for construction safety management.

This paper (or closely related research) has not been published or accepted for publication. It is not under consideration by another journal or by ELSEVIER publications. None of the authors have any competing interests in the manuscript.

If you have questions, please contact me at email: sblee@inu.ac.kr

Sincerely,

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Pedestrian Safety Audits near Construction Sites via Open-Vocabulary Vision-Language Segmentation with Few-Shot Filtering

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Highlights

- Few-shot filtering improves reliability of pseudo mask-label pairs in open-vocabulary models.
- Higher filtering thresholds enhance performance but reduce the number of retained masks.
- Support set size shows minimal effect due to visual homogeneity of sidewalk obstructions.
- Framework processes 4,959 street-view images in under 30 minutes for safety audits.

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Abstract

Ensuring pedestrian safety near construction sites requires frequent audits to detect temporary obstructions, yet current practices are labor-intensive and time-consuming. This study proposes a novel framework that combines open-vocabulary vision-language segmentation with few-shot filtering to automatically generate pseudo-labeled datasets of sidewalk obstructions using street-view images. By integrating Autodistill and a few-shot filtering process, the framework effectively discards unreliable predictions and retains only semantically consistent mask–label pairs. Experimental results show significant improvements in F1 score and MCC compared to baseline models, with especially notable gains for construction domain-specific classes such as fences and traffic cones. The approach also reduces annotation time considerably, enabling rapid dataset generation with minimal human input. As a result, the study contributes a scalable and low-effort approach to auditing construction-related risks to pedestrian safety and offers practical tools for city officials to enhance sidewalk monitoring across diverse urban contexts.

Keywords: Pedestrian Safety, Few-shot Learning, Open-vocabulary Segmentation, Vision-language Model, Pseudo-Label Generation

1. Introduction

Construction projects in urban areas frequently cause disruptions to residents by generating noise, dispersing dust, and creating vibrations that negatively affect the surrounding environment [1-3]. Among these daily disturbances, unauthorized occupation of sidewalks near construction sites by building materials and equipment has recently drawn increased attention [4]. Previous studies have consistently indicated that pedestrians experience discomfort while walking near construction sites due to factors such as narrow sidewalks, uneven surfaces, and the need to navigate around obstructions [5-7]. Other studies have raised concerns that sidewalk obstructions, in the absence of alternative pathways, may force pedestrians to step onto roadways or other unsafe areas not intended for walking, which can result in slips, falls, and traffic-related accidents [8-10].

Despite the growing emphasis on ensuring pedestrian safety during construction activities, regulations aimed at managing the walking environment near construction sites remain limited. For example, although regulations in several cities clearly prohibit the placement of building materials and equipment on sidewalks without a permit, these provisions often lack specified enforcement rules [11,12]. Moreover, systematic tools and guidelines for city inspectors to monitor violations near construction sites are rarely available. As a result, current audit practices typically involve inspectors using checklists to manually assess compliance with pedestrian safety requirements during on-site visits [12,13]. These methods are labor-intensive and time-consuming, often making it difficult to perform frequent inspections across numerous construction sites. Additionally, the subjective nature of manual audits can lead to inconsistencies in evaluating pedestrian environments.

To address these limitations, researchers have begun using street-view images (i.e., panoramic photographs and videos captured at approximately eye level) to audit sidewalk

conditions in urban areas. For example, Steinmetz-Wood et al. [14] developed a Google Street View-based tool for microscale built-environment assessment and reported agreement rates of at least 80 % for most items when virtual audits were compared with field audits. With advancements in computer vision, built-environment features can also be automatically extracted from street-view images, enabling large-scale and objective evaluations of sidewalk conditions. Several studies have employed semantic segmentation approaches – such as SegNet [15], PSPNet [16], DeepLabv3+ [17], and HRNet [18] – to categorize urban-scene elements that facilitate or hinder walking and to quantify the visual characteristics of pedestrian environments.

However, the effectiveness of such computer-vision-based audit models depends on the availability of large, labeled datasets [19,20]. For example, Chen et al. [16] and Li et al. [17] both relied on the Cityscapes dataset, a publicly available collection of high resolution street-view images captured from 50 cities, to train their audit models. This dataset is pre-annotated at the pixel level with 19 common urban scene classes. Another example of a large-scale benchmark for semantic segmentation in street scenes is the Mapillary Vistas dataset, which comprises 25,000 street-view images collected from around the world and annotated into a more diverse set of 66 object classes [20]. Despite the usefulness of existing datasets such as Cityscapes and Mapillary Vistas, they are limited in addressing pedestrian safety issues that arise near construction sites. This limitation stems from the fact that their object classes primarily represent static, permanent features of the urban environment – such as utility poles, benches, vegetation, and bike racks – rather than temporary obstructions like traffic cones, fences, or equipment that may encroach on pedestrian pathways. Additionally, expanding the set of object classes to include temporary obstructions requires considerable manual effort. For example, during the construction of the Mapillary Vistas dataset, image annotation was carried

out by a team of 69 professional annotators, with the average annotation time reported to be approximately 94 minutes per image [20].

Consequently, there is a pressing need for automated methods to identify and annotate the temporary visual features found near construction sites, which are often missing from urban street scene datasets. In the field of computer vision, such challenges have recently been explored through the concept of open-vocabulary segmentation, where models are not limited to a fixed set of labels but instead generate segmentations based on text prompts [21]. One of the promising approaches that adopts the concept of open-vocabulary segmentation is Autodistill, introduced by Roboflow [22]. This open-source framework streamlines the creation of computer vision models by automatically generating training data (i.e., pseudo-labels and corresponding masks) from raw images using text prompts and vision-language models, without requiring any human intervention [22].

This study adapts and extends the Autodistill framework to automatically generate pseudo-labeled datasets of temporary obstructions in street-view images near construction sites. To improve the reliability of the generated pseudo mask-label pairs, the framework incorporates few-shot learning-based filtering process that removes low-quality pseudo-labels. This reduces the need for extensive manual annotation and enhances the feasibility of computer vision-based sidewalk audits near construction sites, ultimately contributing to pedestrian safety. The remainder of this paper is organized as follows. Section 2 introduces open-vocabulary segmentation and its applications in urban and construction domains. Section 3 describes the proposed method integrating Autodistill with few-shot learning. Section 4 presents performance results based on variations in the number of shots and threshold levels. Sections 5 and 6 discuss the findings and conclude the study with directions for future research.

2. Literature Review

2.1. Open-vocabulary semantic segmentation

Recent advances in computer vision have led to the emergence of open-vocabulary semantic segmentation, a technique that enables models to segment and classify objects based on arbitrary text prompts, including classes not present in their original training sets [21,23]. Unlike traditional semantic segmentation methods that require extensive labeled data for each specific class, open-vocabulary approaches leverage large-scale pretrained vision-language models to transfer visual-semantic knowledge extracted from vast image-text datasets to downstream segmentation tasks. This approach significantly reduces the dependency on manually annotated data, allowing more flexible and scalable model deployment.

A prominent example of an open-vocabulary segmentation framework is Autodistill, an open-source tool introduced by Roboflow [22]. In this framework, pretrained vision-language models, referred to as base models, include Segment Anything Model 2 (SAM2) [24] and Florence-2 [25], both of which provide the foundational capability for open-vocabulary understanding and segmentation performance.

Figure 1 illustrates the Autodistill workflow, beginning with caption ontology and an unlabeled image. Caption ontology defines how target objects should be identified and labeled within unlabeled images [26]. Typically, caption ontology employs a JSON format, pairing natural-language prompts with corresponding labels, such as {‘milk bottle’: ‘bottle’}. These prompts explicitly guide the semantic understanding performed by the Florence-2 model.

Using the defined ontology, Florence-2 scans each image to detect objects matching the provided text prompts. As illustrated in Figure 1, Florence-2 identifies objects, such as milk bottles, generating bounding boxes along with appropriate pseudo-labels (e.g., bottle). However, bounding box annotations tend to result in reduced accuracy in segmentation tasks,

as they lack the pixel-level precision necessary to delineate object boundaries accurately [27,28].

Consequently, the bounding boxes generated by Florence-2 are used as inputs to the SAM2, which segments pixels precisely within each box to produce detailed masks delineating object boundaries. For example, given a bounding box containing a milk bottle, SAM2 generates a segmentation mask that accurately captures the object's shape at the pixel level [24]. These annotations are then used to train target models such as YOLOv8 for efficient downstream object detection or segmentation tasks.

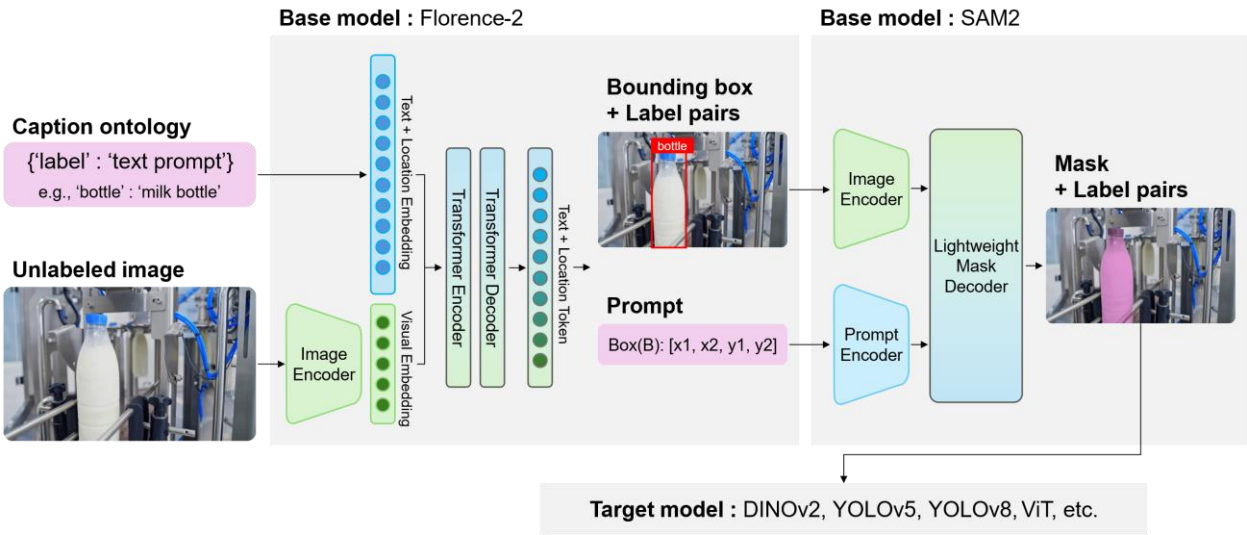

Figure 1. Workflow of the Autodistill framework

2.2. Applications of open-vocabulary frameworks

Several studies have adopted open-vocabulary frameworks in the context of urban planning and construction-related domains. For example, Wu et al. [29] proposed a method for producing land-use classification maps by leveraging street-view images in New York City. Using CLIP as the pretrained vision-language model [30], they explored which types of text prompts – representing different land use categories (e.g., residential, commercial, etc.) – are

most effective in inferring semantic similarity with image features. Liang et al. [31] also utilized CLIP to recognize temporary construction site objects such as rope, trolley, and ladder. The study further compared simple prompts (e.g., trolley) with handcrafted descriptive prompts (e.g., transport and carry heavy loads) and found that the latter significantly improved classification performance. Despite the usefulness of such studies, CLIP-based approaches inherently rely on global image-text similarity and are therefore limited to identifying whether a certain object is present in an image, rather than generating precise bounding boxes or segmentation masks. Furthermore, these studies focused on images with a single dominant object, which may be less effective in real-world urban scenes where multiple objects are present.

Table 1. Overview of prior applications of open-vocabulary vision-language models

Author	Dataset (n^*)	Class	Vision-Language Model	Performance	
				Metric	Score range (%)
Wu et al. [29]	Google Street View images (3398)	Residential, Commercial, Industrial, Greenfield, Recreation, Transportation	CLIP	Degree of matching	39.02 – 66.39
Liang et al. [31]	Photos by authors (90)	Rope, Trolley, Ladder, Trowel, Rebar, Shovel, Helmet, Cone, etc.	CLIP	Accuracy	79.17 – 85.42
Ho et al. [32]	Google Street View images (409)	Door	Grounding DINO**, SAM**,	AP_{50}	48.45 – 78.45
Dong et al. [33]	Photos by authors (4396)	Rock, Gravel, Earth, Packaging, Wood, etc.	Grounded SAM**	MIoU	0.73 – 51.58
Jeoung et al. [34]	Alberta Construction Image dataset (1000)	Dump truck, Excavator, Wheel loader, etc.	Florence-2**, SAM2**,	F1-score	55.4 – 83.3

* *n* indicates the number of images used in the respective study.
** These models are supported as base models in the Autodistill framework.

In a manner similar to the Autodistill framework, Ho et al. [32] proposed a method for estimating the lowest floor elevation of buildings to assess flood risk by segmenting front doors in street-view images. They collected 409 panoramic Google Street View images from flood-prone areas in Texas and applied a combination of SAM and Grounding DINO [35] to perform text-prompt-based segmentation. Dong et al. [33] compared the performance, computational time, and robustness of 17 computer vision models for automated construction and demolition waste sorting. Among them, Grounded SAM was uniquely included as a model that bypasses the need for annotated training data, instead utilizing text prompts to directly generate segmentation masks. In another related study, Jeoung et al. [34] developed a method to recognize construction equipment tasks from video frames by integrating Florence-2 and SAM2. In their framework, Florence-2 was employed as the object detection module, while SAM2 was used for the object tracking module, enabling equipment task recognition over time in dynamic construction scenes.

While prior studies have demonstrated the feasibility of open-vocabulary frameworks, they have also raised concerns about the reliability of pseudo mask-label pairs. As shown in Table 1, the overall performance scores of the models were observed to be relatively low, often averaging below 60%. In this regard, some researchers have noted that segmentation accuracy varies significantly depending on the phrasing of text prompts [29,31,32,34]. This variation suggests that current image-text matching approaches may be unstable. The study that leveraged Grounded SAM without any fine-tuning or post-processing reported that its low accuracy may be attributed to the lack of construction-specific domain knowledge in general-purpose vision-language models [33].

2.3. Challenges and few-shot solutions

Through state-of-the-art literature reviews, it was found that pseudo mask-label pairs often contain fragmented regions or inaccurate mask-label pairs, which can degrade the performance of computer vision models. Some existing studies have attempted to address this issue on the model side through pre-processing strategies such as model fine-tuning or similarity-cache [31,36]. While these approaches offer the advantage of directly improving segmentation results through inference-side adaptation, they still inherit errors from the original pseudo masks.

Therefore, this study proposes an additional filtering stage to screen the generated mask-label pairs before they are used as training data for downstream models. As a complementary alternative to fully zero-shot approaches, this study adopts few-shot learning, training models with very limited labeled examples (shots) per class [37]. The basic idea of incorporating few-shot learning into Autodistill's workflow is to compare automatically generated data against a small number of carefully selected representative examples. For instance, employing a few-shot learning approach with 5-shot, 10-shot, or 30-shot configurations allows this study to retain high-quality samples with greater confidence, while discarding pseudo-labels that fail to meet a defined quality threshold. The proposed post-processing approach improves data quality without relying on variations in text prompts and can be integrated with any downstream computer vision model. However, as strict filtering inevitably reduces the total number of generated masks, this study also investigates the trade-off between dataset size and annotation quality.

3. Method

3.1. Data collection

Video footage was collected from sidewalks near construction sites in the Seoul Metropolitan Area of South Korea, which includes Seoul, Incheon, and Gyeonggi Province. The footage was recorded using a smartphone (2.8K resolution), with the camera held at approximately eye level to simulate a pedestrian's point of view. During the data collection period between March 15 and April 5, 2024, a total of 85 minutes of footage was recorded across 38 different sites. Street-view images were ultimately extracted at a rate of one frame per 60 frames, resulting in a dataset of 4,959 images used in this study. Figure 2 shows sample street-view images captured near construction sites. As shown in the figure, videos were recorded without pedestrians in the frame to focus on visible physical features of the pedestrian environment, including obstructions resulting from construction activities.

Figure 2. Example street-view images extracted from video footage collected for this study

3.2. Class selection

Four object classes were selected as targets for open-vocabulary segmentation, based on their frequent occurrence in the collected dataset and their documented relevance to pedestrian safety. These classes represent both potential hazards and protective elements in pedestrian environments affected by nearby construction activities. For example, parked cars have been widely cited as major obstructions to pedestrian movement, particularly when they encroach on sidewalks near construction sites. Likewise, poorly maintained sidewalks can heighten the risk of slips, trips, and other pedestrian accidents. In contrast, fences and traffic cones are commonly used to separate pedestrians from construction sites, although cones often fail to provide sufficient physical protection. A detailed summary of each class and its relevance to pedestrian safety is provided in Table 2.

Table 2. Selected object classes and their relevance to pedestrian safety

Class	Index	Relevance to pedestrian safety	References
Fence	C1	Fences help define boundaries around construction sites and are associated with an increased perception of safety among pedestrians.	[7,10,12]
Sidewalk	C2	The condition of sidewalks (e.g., continuity and surface evenness) affects pedestrian walkability and risk exposure.	[7-9,12]
Parked car	C3	Parked cars frequently occupy sidewalks near construction sites, which may obstruct pedestrian movement and increase collision risk.	[5,6,8,9,11,12, 38]
Traffic cone	C4	Traffic cones are used to guide or separate pedestrian flow but are often insufficient as a physical safety barrier.	[12,39]

3.3. Caption ontology

Careful prompt engineering enables pretrained vision-language models to segment precisely defined and semantically consistent object classes. A straightforward way to generate text prompts is to use one or two simple words that refer to the target object, such as “fence”. Although this approach offers clarity, it often lacks the semantic richness needed to effectively guide models that interpret prompts as sequences of discrete tokens [25]. Several studies have therefore demonstrated that descriptive prompts incorporating additional visual attributes such as color, texture, and function consistently outperformed simpler labels (e.g., “a blue, fabric like barrier near a construction site”) [31,34]. With respect to expression style, reasoning-based prompts have been reported to be particularly effective compared with declarative statements, because many vision-language models are pretrained to answer natural language questions about visual content [40,41].

Table 3. Class specific reasoning style prompts used in this study

Class	Prompt
Fence	What blue, fabric-like barrier, explicitly designed to shield or guide pedestrians near construction sites, is visible in this image?
Sidewalk	What type of pedestrian pathway, found alongside roads or within construction sites, composed of materials like nonwoven fabric, sand, bricks, or asphalt, is visible in this image?
Parked car	What motorized vehicle, designed for passenger or cargo transport, commonly seen in urban or road environments, is visible in this image?
Traffic cone	What small, brightly colored cone-shaped object, specifically designed to redirect traffic or highlight construction hazards, is visible in this image?

Building on these observations, this study established the following prompt generation strategy. Because manually crafted text prompts that rely on human intuition often fail to achieve optimal performance [41-43], this study leveraged GPT-4o to generate reasoning prompts. Each image was first converted into a natural language description focusing on salient visual features using ChatGPT-4 in a text only setting. Based on these descriptions, the model was then prompted to generate reasoning style questions that begin with "What" and end with "is visible in this image?". Table 3 presents the resulting prompts for each class. Lastly, the automatically generated prompts were qualitatively reviewed by three independent researchers to confirm clarity and appropriateness for visual reasoning.

3.4. Annotation generation with Autodistill

After collecting unlabeled street-view images and defining both the target object classes and their associated caption ontology, the Autodistill framework was implemented to generate pseudo-labels for downstream training. In this study, Florence-2 and SAM2 were selected as base models because prior studies demonstrated that this specific combination achieved reasonable performance among open-vocabulary segmentation models [34]. Following the workflow illustrated in Fig. 1, the reasoning-based prompts listed in Table 3 were first reformatted into a JSON format. These prompts were then used as input to the Florence-2 base model (0.23B parameters) [25]. Within the model, the encoder jointly embedded the image and prompt representations to enable cross-modal semantic understanding, while the decoder generated bounding box coordinates that localized image regions corresponding to each prompt. Bounding boxes were represented as 2D coordinates of the top-left and bottom-right corners (i.e., x_{min}, y_{min}, x_{max}, y_{max}).

In the next stage, the SAM2 (Hiera-B+) was employed to generate pixel-level segmentation masks [24]. Using the bounding box coordinates predicted by Florence-2 as box prompts, SAM2 restricted segmentation to the prompted regions rather than the entire image. The fusion of image and prompt features was performed in the lightweight mask decoder, and the resulting pixel-wise probabilities of object presence were then converted into binary segmentation masks by applying a threshold of 0.5, which is the default setting. Only masks greater than or equal to the threshold were retained, and non-maximum suppression was applied to remove duplicate predictions. All stages were executed on a local machine equipped with an Intel Core i9 CPU, 128 GB RAM, and an NVIDIA RTX 4090 GPU.

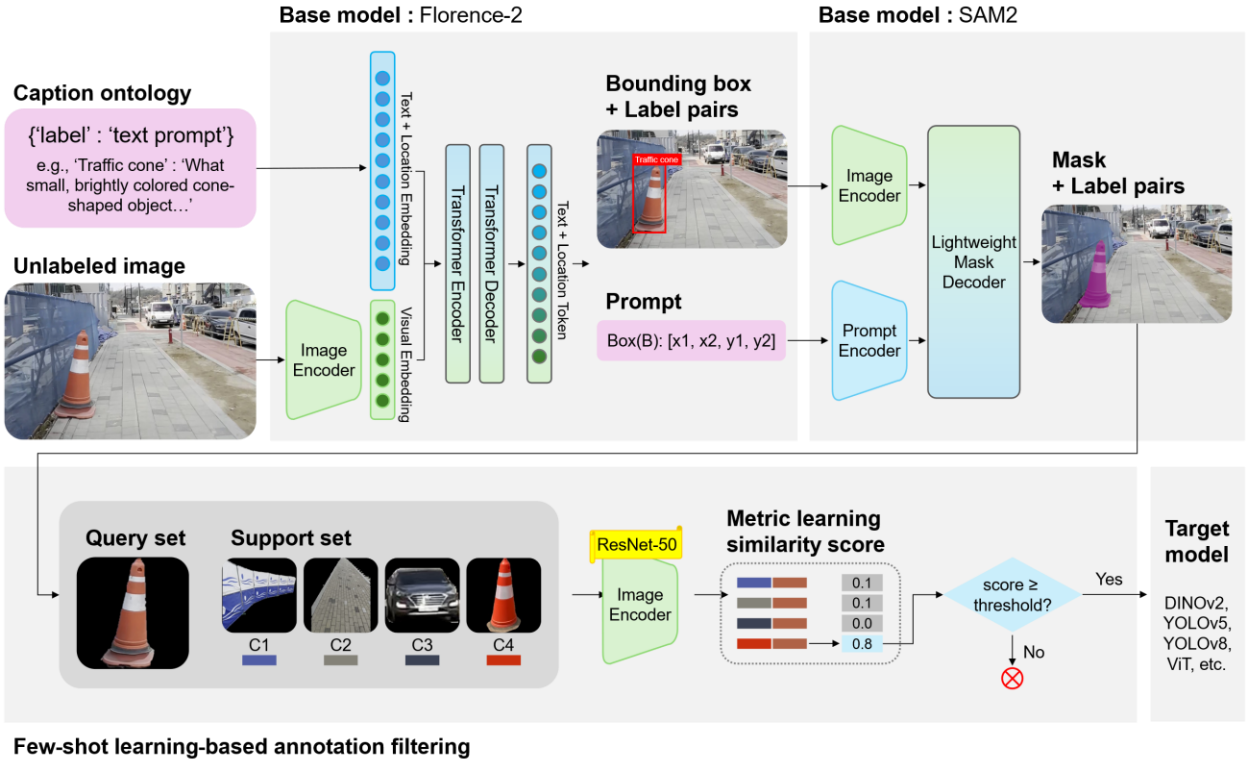

Figure 3. Pseudo mask-labels generation workflow with few-shot learning-based filtering

3.5. Few-shot-based annotation filtering

To enhance the reliability of pseudo mask-labels generated by Autodistill, this study added few-shot learning-based annotation filtering stage. As illustrated in Fig. 3, both the query and support sets were constructed by preprocessing object masks generated during the SAM2 segmentation stage. Each object mask was tightly fitted to its outermost boundary, cropped, and resized to a fixed resolution of 224×224 pixels. Background pixels outside the mask were removed by setting them to transparent, ensuring that only the object region remained visible. In addition, to construct the support set – which consists of pseudo-labeled object masks verified for use as class-wise references – candidate samples were randomly suggested by the model, from which the user selected representative instances. Figure 4 shows a sample of support set masks selected for each class under the 5-shot setting.

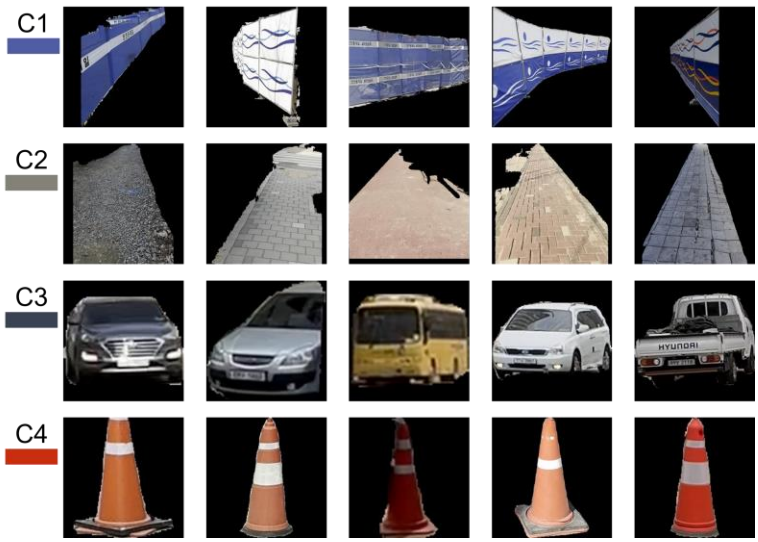

Figure 4. Examples of class-wise support set (5-shot per class)

Once the support and query sets were prepared, feature embeddings were extracted using an image encoder built on ResNet-50 as its backbone [44]. This study specifically adopted a metric-learning-based method, in which query and support masks were embedded into a shared feature space, and their pairwise similarity was used to infer whether they belonged to the same class [37,45]. Since the classification task followed a 4-way K-shot setting, similarity scores between the query and each of the four support classes were computed and normalized to sum to 1. The class with the highest score was then selected as the tentative pseudo label. If this score exceeded a predefined threshold, the corresponding pseudo mask-label pair was retained for downstream use; otherwise, it was discarded. This threshold acted as a confidence filter, and values ranging from 0.5 to 0.8 were tested during performance evaluation.

3.6. Evaluation metrics

Open-vocabulary segmentation inherently involves challenges related to imbalanced class distributions [33,45,46]. The proposed method is similarly affected by this imbalance, as both the query and support sets are derived from the natural class distribution in street-view images. For example, in our dataset, while sidewalks appear in the majority of collected images, traffic cones are observed less frequently. In such settings, it is important to use evaluation metrics that fairly assess the model's performance on minority classes [47].

Therefore, this study adopted four evaluation metrics widely used in multi-class classification with imbalanced datasets: macro balanced accuracy, macro F1 score, macro Matthews correlation coefficient (MCC), and macro fall-out. These metrics were derived from the confusion matrix, which summarizes prediction outcomes using four components: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). To assess the model's overall performance,

each metric was first computed for individual classes using a one-vs-rest strategy and then macro-averaged across all classes.

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i} \tag{1}$$

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \tag{2}$$

where $i \in \{1, \ldots, n\}$ indicates the class index, and n is the total number of classes.

The first metric, macro balanced accuracy, compensates for class imbalance by averaging both sensitivity (recall) and specificity across all classes [48]:

$$\text{Macro Balanced Accuracy} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{2} \left(\text{Recall}_i + \frac{TN_i}{TN_i + FP_i} \right) \tag{3}$$

Macro F1 Score is computed by first calculating the F1 score for each class independently and then taking their unweighted mean [48]. This ensures that all classes are given equal importance, regardless of their frequency in the dataset. If both precision and recall are zero for a given class, its F1 score is defined as zero:

$$\text{Macro F1-Score} = \frac{1}{n} \sum_{i=1}^{n} \frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{4}$$

Macro MCC incorporates all four elements of the confusion matrix and is computed per class, then averaged across all classes. It ranges from −1 to +1, where +1 indicates perfect prediction, 0 corresponds to random guessing, and −1 indicates complete disagreement between the predictions and the ground truth [47]:

$$\text{Macro MCC} = \frac{1}{n} \sum_{i=1}^{n} \frac{TP_i \times TN_i - FP_i \times FN_i}{\sqrt{(TP_i + FP_i)(TP_i + FN_i)(TN_i + FP_i)(TN_i + FN_i)}} \tag{5}$$

Lastly, macro fall-out, also known as the false positive rate (FPR), measures the proportion of negative samples that are incorrectly predicted as positive [49]. The fall-out for a given class is defined only when the sum of false positives and true negatives is greater than zero, to avoid division by zero:

$$\text{Macro Fall-out} = \frac{1}{n}\sum_{i=1}^{n}\frac{FP_i}{FP_i + TN_i} \tag{6}$$

4. Results

4.1. Initial annotations generated by Autodistill

Prior to applying few-shot learning-based annotation filtering process, Autodistill produced an initial set of 16,061 segmentation masks from 4,959 raw street-view images. Each segmentation mask was automatically pseudo-labeled as one of four predefined object classes. To evaluate the accuracy of these initial annotations, all generated masks were manually reviewed by the authors to establish ground-truth labels. Table 4 summarizes the manual validation of all segmentation masks initially generated by Autodistill. Out of 16,061 masks, only 9,412 masks (58.6%) were confirmed to belong within the four target classes, while the remaining 6,649 masks (41.4%) were manually marked as “others”. The “others” category frequently included humans and often corresponded to ambiguous or unidentifiable objects.

Table 4. Summary of manually validated initial annotations

True class	Within target classes (C1~C4)	Manually marked as "others"	Total
Fence (C1)	534	553	1087
Sidewalk (C2)	3286	2183	5469
Parked car (C3)	4214	1241	5455
Traffic cone (C4)	1378	2672	4050
Total	9412	6649	16061

To establish a reliable ground-truth set of mask-label pairs, the 9,412 masks identified as belonging within the four target classes were further classified according to their actual classes. Table 5 shows the confusion matrix comparing Autodistill predictions and manually established ground-truth labels. For example, among the 534 masks identified as fences (C1), only 45 masks were accurately classified as fences. The remaining masks were predominantly misclassified as sidewalks (C2; 478 cases) or parked cars (C3; 11 cases). In contrast, sidewalk (C2) annotations exhibited a high match rate within the target classes, with 3,284 out of 3,286 masks correctly classified.

Table 5. Confusion matrix for initial annotations within target classes

True class	Predicted class				
	C1	C2	C3	C4	Total
Fence (C1)	45	478	11	0	534
Sidewalk (C2)	1	3284	1	0	3286
Parked car (C3)	259	0	3955	0	4214
Traffic cone (C4)	61	0	4	1313	1378
Total (Ground-truth)	366	3762	3971	1313	9412

Table 6 presents the performance evaluation metrics computed for each class. Masks manually marked as "others" (Table 4) were considered incorrect predictions when the model assigned them to any of the predefined classes. In a binary evaluation for a given class, such cases were counted as false positives when predicted as that class, and as true negatives otherwise. As shown in Table 6, the performance varied substantially across object classes. Parked cars (C3) consistently outperformed other categories across all metrics, achieving the highest balanced accuracy (93.60%), F1 score (83.92%), and MCC (0.79). Sidewalks (C2) also demonstrated strong performance, particularly in terms of balanced accuracy (84.76%) and F1 score (71.15%).

Table 6. Class-wise performance metrics of initial annotations

Class	Balanced accuracy (%)	F1 (%)	MCC	Fall-out (%)
Fence (C1)	52.83	6.19	0.03	6.64
Sidewalk (C2)	84.76	71.15	0.62	17.77
Parked car (C3)	93.60	83.92	0.79	12.41
Traffic cone (C4)	90.72	48.97	0.51	18.56
Macro average	80.48	52.56	0.49	13.84

In contrast, fences (C1) showed the poorest performance, with extremely low F1 score (6.19%) and MCC (0.03). Traffic cones (C4) achieved relatively high balanced accuracy (90.72%) but suffered from low F1 (48.97%) and MCC (0.51). These results suggest that the conventional model performs well on general urban objects such as parked cars and sidewalks but struggles to generalize to less frequent construction-related objects such as fences and traffic cones. Macro-level scores, computed by averaging class-wise metrics equally across all four classes, are also

included at the bottom of Table 6 to summarize the overall model performance. These scores demonstrate the need to improve F1 and MCC for construction-related classes.

4.2. Filtered annotations by few-shot learning

To assess the impact of the proposed few-shot learning-based filtering process on annotation quality, the following tables present results obtained under the 10-shot, threshold = 0.40 configuration. As shown in Table 7, the initial 16,061 masks generated by Autodistill were reduced to 10,072 after applying the filtering process, of which 7,919 masks (78.6%) were manually validated as belonging to one of the four target classes. Compared to the initial annotations (Table 4), the number of masks labeled as "others" was substantially reduced from 6,649 to 2,153, showing that the filtering process effectively removed a large portion of misclassified or ambiguous labeled objects. In particular, fences (C1) and parked cars (C3) showed the largest relative reductions in "others" (94.4% and 90.6%, respectively), indicating that the filtering process was especially effective in improving precision for these classes.

Table 7. Summary of manually validated annotations after filtering

True class	Within target classes (C1~C4)	Manually marked as "others"	Total
Fence (C1)	286	31	317
Sidewalk (C2)	3351	1258	4609
Parked car (C3)	3139	116	3255
Traffic cone (C4)	1143	748	1891
Total	7919	2153	10072

Note: Values were evaluated under a fixed 10-shot setting with a threshold of 0.40.

The resulting confusion matrix after filtering is shown in Table 8. Compared to the pre-filtering stage (Table 5), misclassifications were almost entirely eliminated. For example, fences (C1), which were previously misclassified as sidewalks (C2) in 478 cases, were now correctly identified in 285 out of 286 instances. Sidewalks (C2) and parked cars (C3) also achieved near-perfect agreement with their ground-truth labels, with only one misclassification each. Traffic cones (C4) exhibited perfect precision, with no false positives and only one false negative, further indicating the substantial improvement in label quality achieved through the filtering process.

Table 8. Confusion matrix after filtering within target classes

True class	Predicted class				
	C1	C2	C3	C4	Total
Fence (C1)	285	1	0	0	286
Sidewalk (C2)	1	3350	0	0	3351
Parked car (C3)	0	0	3138	1	3139
Traffic cone (C4)	0	0	0	1143	1143
Total (Ground-truth)	286	3351	3138	1144	7919

Note: Values were evaluated under a fixed 10-shot setting with a threshold of 0.40.

Table 9 presents the class-wise performance metrics after few-shot filtering process. As in the initial annotations case, masks manually marked as “others” (Table 7) were handled in the same way when calculating each metric. Compared to the initial annotations (Table 6), all classes showed substantial improvements across balanced accuracy, F1 score, and MCC, while the average fall-out rate decreased by nearly half. From a class-wise perspective, parked cars (C3) still achieved the highest performance across all metrics, with balanced accuracy of 99.16%, F1 score of 98.17%, and MCC of 0.97. Fences (C1), which previously had the lowest F1 score (6.19%) and

MCC (0.03), improved dramatically to 94.53% and 0.94, respectively. Traffic cones (C4) also showed notable gains in F1 and MCC, reaching 75.32% and 0.74, respectively. Sidewalks (C2) maintained strong overall performance but recorded the highest fall-out rate (18.73%) among all classes. The macro-average scores represent a substantial overall improvement from the pre-filtering stage.

Table 9. Class-wise performance metrics after filtering

Class	Balanced accuracy (%)	F1 (%)	MCC	Fall-out (%)
Fence (C1)	99.66	94.53	0.94	0.33
Sidewalk (C2)	90.62	84.17	0.77	18.73
Parked car (C3)	99.16	98.17	0.97	1.69
Traffic cone (C4)	95.77	75.32	0.74	8.38
Macro average	96.30	88.05	0.86	7.28

Note: Values were evaluated under a fixed 10-shot setting with a threshold of 0.40.

To further investigate how the performance of the proposed framework varies with the support set size and the filtering threshold, Tables 10 and 11 summarize the macro-level performance metrics under different configurations. Table 10 presents the results when the support set size is varied under a fixed filtering threshold of 0.40. As the support set size increased, balanced accuracy, F1 score, and MCC showed a slight but consistent decrease, while fall-out rates increased. However, the magnitude of these changes was small, with all configurations maintaining balanced accuracy above 96%, F1 above 87%, and MCC above 0.85. This indicates that the framework performs stably across a wide range of support set sizes, and that even relatively small support sets can achieve performance comparable to larger ones under the given threshold.

Table 10. Macro performance metrics by support set size

Shot	Balanced accuracy (%)	F1 (%)	MCC	Fall-out (%)
1-shot	96.93	89.87	0.89	3.59
5-shot	96.68	89.05	0.87	6.33
10-shot	96.30	88.05	0.86	7.28
30-shot	96.18	87.59	0.85	7.54

Note: Shot values were evaluated under a fixed threshold of 0.4.

Table 11 presents the macro-level performance metrics when varying the classification threshold under a fixed 10-shot setting. As the threshold increased, all performance metrics improved because it retained only the most confident predictions and thereby enhanced both precision and recall among the remaining samples. Notably, for filtering thresholds from 0.30 through 0.45, both F1 score and fall-out improved substantially, mainly due to a sharp drop in false positives while most true positives were retained. At around 0.50, nearly all metrics reached peak performance.

Table 11. Macro performance metrics by filtering threshold

Threshold	Balanced accuracy (%)	F1 (%)	MCC	Fall-out (%)
0.30	92.60	72.85	0.71	11.06
0.35	94.94	80.90	0.79	9.33
0.40	96.30	88.05	0.86	7.28
0.45	97.90	94.09	0.92	4.09
0.50	99.47	98.30	0.98	1.06
0.55	99.84	99.46	0.99	0.32

Note: Threshold values were evaluated under a fixed 10-shot setting.

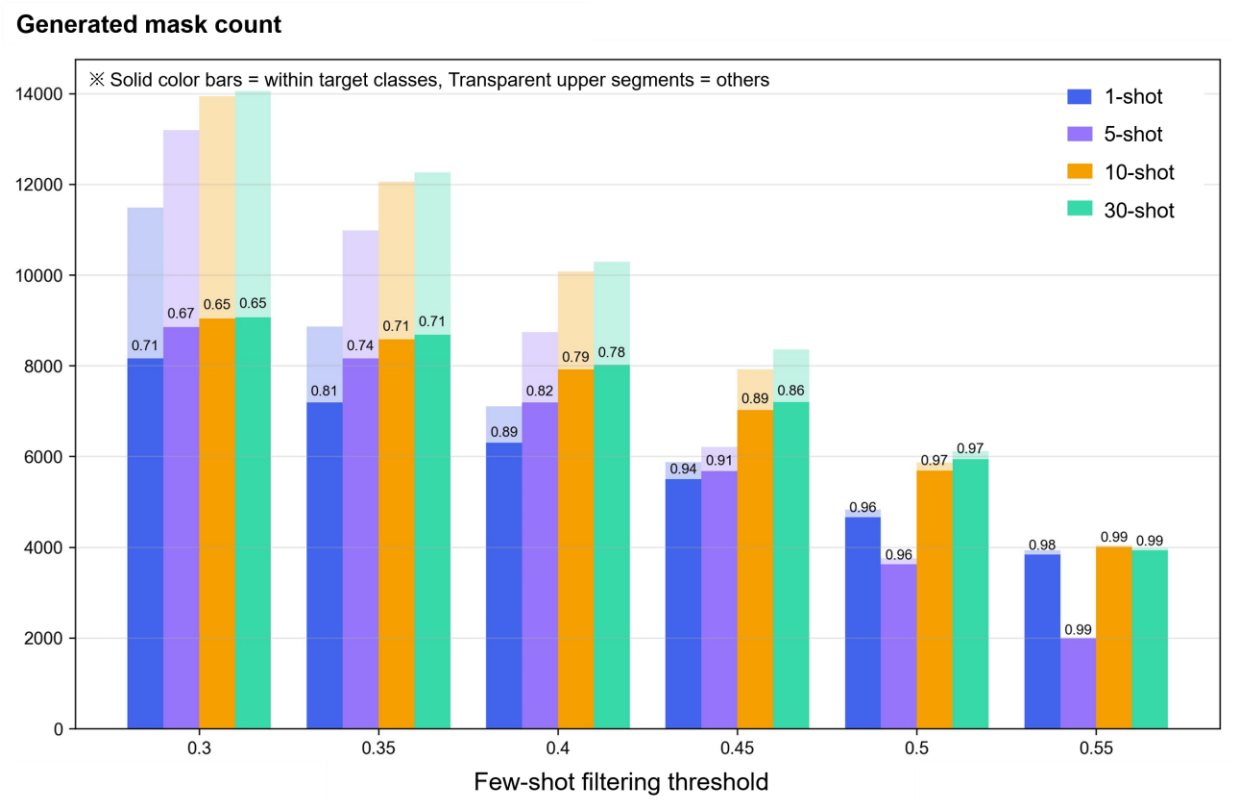

Figure 5. Effect of threshold and support set size on mask generation

While higher thresholds improve performance metrics, they inevitably filter out a larger portion of predictions, thereby reducing the total number of masks generated. To examine this trade-off between annotation quality and dataset size, Fig. 5 illustrates the combined effect of few-shot filtering thresholds and support set size on mask generation. Each group of bars corresponds to a threshold value, with colors indicating the number of support shots. Solid color bars represent masks retained within the four target classes, and transparent upper segments denote masks classified as "others." As the threshold increases, both the total mask count and the proportion of "others" decline markedly. For example, at a threshold of 0.30, the proportion of solid bars ranges from 65% to 71%, whereas at thresholds of 0.50 and above it reaches about 96%. This indicates that higher thresholds increase label precision by filtering out low-confidence predictions.

However, in the context of generating pseudo mask-label pairs for downstream target model training, this comes at the cost of reducing dataset size, requiring a balance between annotation quality and quantity.

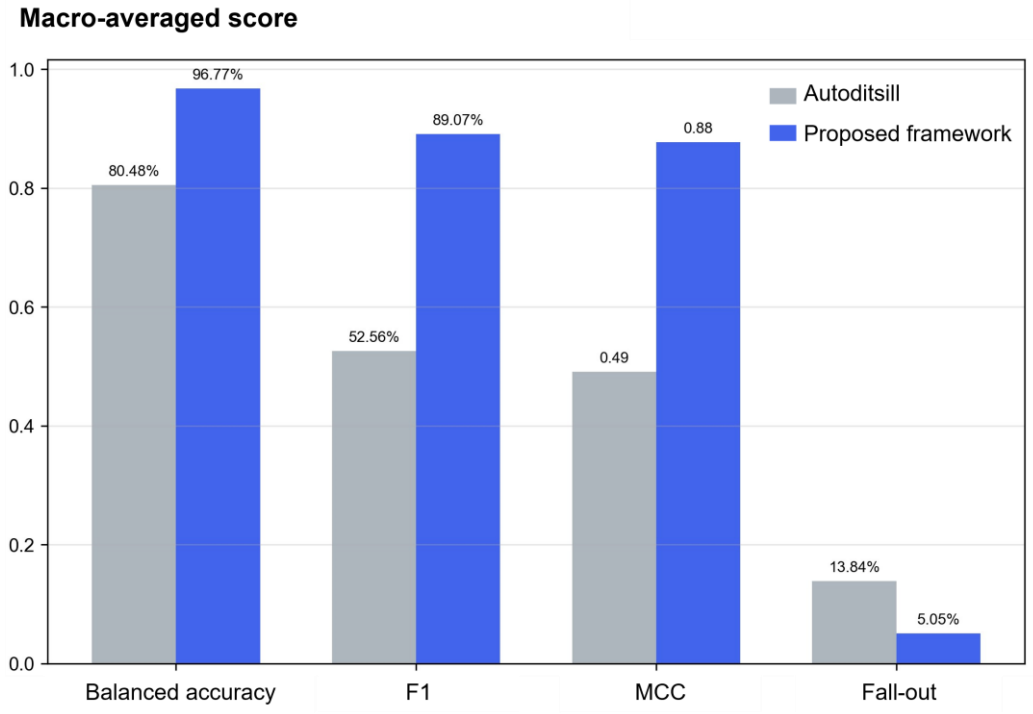

Figure 6. Performance comparison between Autodistill and the proposed framework

4.3. Performance comparison and processing time

Fig. 6 compares macro-averaged performance scores across all four metrics between the Autodistill baseline and the proposed framework. For the proposed framework, values were averaged over all 24 experimental configurations (six thresholds × four support shot sizes). Compared to the Autodistill baseline, the proposed method achieved substantial gains in balanced accuracy (80.48% → 96.77%, +16.29%), F1 score (52.56% → 89.07%, +36.51%), and MCC (0.49

→ 0.88, +0.39), along with a marked reduction in fall-out (13.84% → 5.05%, -8.79%). These results demonstrate the potential of text-prompt-based open-vocabulary segmentation models for generating reliable pseudo-labeled datasets and confirm that the proposed framework substantially improves upon the original Autodistill approach in creating high-quality training data for downstream computer vision tasks.

Although human-annotated training datasets typically achieve near-perfect reliability, the proposed method attains a slightly lower macro-averaged performance of approximately 90% across all evaluation metrics. However, this limitation is offset by the substantial time savings achieved in generating large-scale pseudo-labeled datasets. As shown in Table 12, generating the initial set of pseudo-labels with Autodistill takes only 180 seconds, followed by few-shot filtering stage that, even when executed across all 24 configurations, completes in just under 30 minutes. Considering that the proposed framework can produce approximately 4,000 to 14,000 segmentation masks within this timeframe, these results demonstrate its practicality and scalability for constructing training datasets in downstream computer vision tasks.

Table 12. Time required for each stage of the proposed model

Stage	Time(sec.)
Autodistill	180
Few-shot filtering*	1796

* Note: Few-shot filtering time is the cumulative processing time for 24 configurations, combining six threshold settings (0.30-0.55) and four support shot sizes (1, 3, 5, and 10).

5. Discussion

5.1. Performance gains from few-shot filtering

This study incorporated few-shot filtering into an open-vocabulary vision-language segmentation framework to produce pseudo-labeled datasets of construction-related objects that pose risks to pedestrian safety in street-view images. This approach enabled the model to retain only masks with high semantic consistency to the target classes while filtering out low confidence predictions and achieved consistent macro-averaged performance gains across all metrics compared to the baseline framework. These gains were particularly strong for F1 score and MCC, indicating that the filtering process substantially reduced the number of false positives while retaining the majority of true positives. This is especially important in the present context, where the goal is not to detect every possible target object in street-view images, but rather to generate a high-precision set of pseudo mask-label pairs before training downstream computer vision models.

At the class level, the most dramatic gains were observed for fences and traffic cones, which initially suffered from very low reliability. This improvement shows that the proposed filtering strategy is particularly effective at recognizing domain-specific urban elements critical to construction safety but often overlooked in the pretraining of vision–language models. It addresses a key limitation of prior open-vocabulary segmentation studies, which have struggled with low pseudo-label reliability [29,31-33]. By incorporating visual similarity scores from the support set, the framework also reduces dependence on precise text prompt wording. Even when prompts are ambiguous or lack domain-specific detail, the support set provides complementary visual context that guides the selection of the most semantically consistent masks, enabling more reliable pseudo-label generation for construction-relevant urban elements described in varied or imprecise language.

5.2. Impact of threshold and support set size

This study evaluated 24 combinations of threshold and support set size to examine their effects on pseudo mask-label pair generation performance. In terms of metrics alone, increasing the threshold consistently improved balanced accuracy, F1 score, and MCC, while reducing fall-out. Pronounced improvements were observed up to a threshold of 0.50, beyond which the effects began to plateau. This trend aligns with prior studies that reported similar optimal ranges [50]. However, as shown in Fig. 5, higher thresholds inevitably reduced the total number of retained masks. This reflects a trade-off between annotation quality and dataset size, a pattern that aligns with previous studies reporting similar tensions in pseudo-label generation frameworks [51,52]. Therefore, depending on the model requirements, one may opt to lower the threshold to prioritize dataset size or raise it to emphasize label accuracy. Although some studies have suggested that larger but noisier pseudo-labeled datasets may outperform smaller, high-precision ones in certain downstream tasks [53], this finding remains context dependent and warrants further exploration.

Regarding support set size, performance metrics exhibited a slight decline from 1-shot to 30-shot, though the differences were minimal and not practically significant. This outcome may be attributed to the visual homogeneity of the target classes. For example, objects such as fences and traffic cones tend to exhibit consistent visual characteristics across different scenes. This consistency was further supported by the data collection setup, in which street-view images were captured at a pedestrian's eye level along sidewalks. Since most target objects were recorded from similar viewpoints and distances, even larger support set sizes may have provided limited additional visual cues for the filtering process.

5.3. Practical implications

The proposed framework offers several practical benefits for auditing pedestrian environments near construction sites. One of the key advantages is the substantial reduction in annotation time. As noted earlier, creating large-scale datasets such as Mapillary Vistas required teams of professional annotators and approximately 94 minutes per image [20]. Even in simpler contexts, a prior study reported that manually annotating 2,400 images required approximately 20 hours, assuming the work is done by a single individual without breaks [26]. In contrast, the proposed framework automatically generated over 10,000 segmentation masks from 4,959 street-view images with minimal human input. The entire pseudo-label generation and filtering process was completed in under 30 minutes, representing only a fraction of the time required for manual annotation in comparable studies.

This time efficiency addresses longstanding challenges in pedestrian safety auditing, which currently relies on labor-intensive field visits and checklist-based assessments. Moreover, the proposed framework lowers the barrier to generating domain-specific training data, as city officials can easily adapt the system to various types of sidewalk obstructions by simply modifying the text prompt and preparing a representative support set. Ultimately, this approach enables a more proactive, scalable, and objective system for ensuring pedestrian safety in construction-affected urban environments.

6. Conclusion

This study presented a novel framework that combines Autodistill and few-shot filtering to automatically generate pseudo-labeled datasets of temporary obstructions near construction sites. The primary objective was to address the lack of construction-specific annotations in existing street-view segmentation datasets, enabling the generation of high-precision mask–label pairs with minimal human input. The proposed method achieved consistent performance improvements across multiple evaluation metrics, particularly F1 score and MCC, by effectively reducing false positives while retaining true positives. The study observed that the framework achieved optimal performance with a filtering threshold around 0.50, while support set size had minimal impact due to the visual consistency of target objects in the dataset. In practical terms, the proposed method dramatically reduces annotation time and lowers the barrier to generating domain-specific training data. It allows city officials to quickly adapt the model to various construction-related obstructions by modifying only the text prompt and support set, enabling more scalable and objective audits of pedestrian environments.

Nevertheless, several limitations should be noted. First, while human-annotated datasets can achieve near-perfect reliability, the proposed framework does not fully match that level of precision. Although this shortfall may be offset by the substantial gains in annotation speed and scalability, it remains important to examine how the use of such automatically generated labels affects the performance of downstream computer vision models. Future studies should further investigate this trade-off, especially in contexts where label accuracy is critical. Second, the support set images were drawn from a narrow range of perspectives (i.e., pedestrian eye level), which may yield suboptimal results when applied to obstructions captured from different angles or elevations. Finally, the study evaluated only a fixed set of text prompts. Therefore, future studies

should examine generalizability by incorporating more diverse image capture methods and investigating the effects of prompt variation to improve flexibility across urban contexts.

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Declaration of Interest Statement

☐ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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